Stochastic Processes 2021/22 Semester 1

Extension Questions 4

These questions are an extension to Tutorial 4, intended for personal revision only. Answers are available for all questions, but worked solutions are only available for some.

1. Consider the Markov chain $\{X_n\}_{n\geq 0}$ taking values in the state space $S = \{1, 2, 3, 4\}$ with transition matrix

$$P = \begin{pmatrix} 1/3 & 0 & 2/3 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 1/3 & 1/3 & 0 & 1/3 \\ 0 & 0 & 1/2 & 1/2 \end{pmatrix}.$$

   (a) Verify that all states form a single, necessarily closed, class, and find the unique stationary distribution for the chain.

   (b) Suppose that $\mathbf{P}(X_0 = i) = 1/4$ for $i = 1, 2, 3, 4$. Find (again) the distributions of the random variables X_1, X_2, X_3 and comment on the rate of convergence to the stationary distribution.

2. Consider the Markov chain $\{X_n\}_{n\geq 0}$ taking values in the state space $S = \{1, 2\}$ with transition matrix

$$P = \begin{pmatrix} 5/8 & 3/8 \\ 1/2 & 1/2 \end{pmatrix}.$$

   (a) Show by induction that

$$P^n = \begin{pmatrix} 4/7 & 3/7 \\ 4/7 & 3/7 \end{pmatrix} + \left(\frac{1}{8}\right)^n \begin{pmatrix} 3/7 & -3/7 \\ -4/7 & 4/7 \end{pmatrix}$$

   (b) Find the stationary distribution $\boldsymbol{\pi}$ associated with this chain and verify that, for all i, j, we have $\lim_{n\to\infty} p_{ij}^{(n)} = \pi_j$.

3. A no claims discount system under a motor policy has five levels of discount: 0%, 25%, 40%, 60% and 75%. Policy holders change level of discount depending on the number of claims they make each year, according to the following table

	Number of Claims		
Current Level	0	1	>2
0%	40%	0%	0%
25%	40%	25%	0%
40%	60%	40%	0%
60%	75%	60%	25%
75%	75%	75%	25%

   The number of claims each year for each policy holder is assumed to have a Poisson distribution with parameter 0.1.

   (a) For 10,000 policyholders at 0% discount at time 0, estimate the expected numbers at each discount level at times 1 and 2, assuming no exits.

   (b) For 10,000 policyholders, estimate the expected numbers at each discount level once stability has been achieved, again assuming no exits.

4. A PhD Student has some unknown (but finite) number of pairs of socks, labelled $1, 2, 3, \ldots, k$. Each day he chooses at random a pair of socks to wear from those he was not wearing the previous day. Let X_n be the label of the socks he wears on day n.

   (a) Explain why $\{X_n\}_{n\geq 1}$ is a Markov chain.

   (b) Show that the transition matrix P may be written

   $$P = \frac{k}{k-1}E - \frac{1}{k-1}I$$

   where E is the $k \times k$ matrix with every element equal to $\frac{1}{k}$ and I is the $k \times k$ identity matrix.

   (c) Prove, by induction on n, that

   $$P^n = \left(1 - \left(-\frac{1}{k-1}\right)^n\right) E + \left(-\frac{1}{k-1}\right)^n I$$

   (d) Find the stationary distribution $\boldsymbol{\pi}$ of the chain, and verify that $\lim_{n\to\infty} p_{ij}^{(n)} = \pi_j$ for all i, j.

   (e) Let $f_{ii}(n)$ be the probability that the next time he wears pair of socks i is in n days time, given that he is wearing them today. Find an expression for $f_{ii}(n)$ and hence find the mean number of days between successive wearings of pair of socks i.

5. Eve has a simple filing system on her desk, a pile of papers. When she needs a document she looks through the pile and after using it places it on the top of the pile. Consider the situation where there are only 3 documents, labelled 1,2,3. The probability she needs document i next is p_i, independent of the previous documents needed.

   (a) Let α_n be the probability that the documents are in the order 1,2,3 (document 1 on the top, 3 on the bottom). Show irrespective of the initial arrangement, α_n converges to a limit as $n \to \infty$, and determine the limit.

   (b) Find the long run proportion of time document 2 is on the top of the pile.